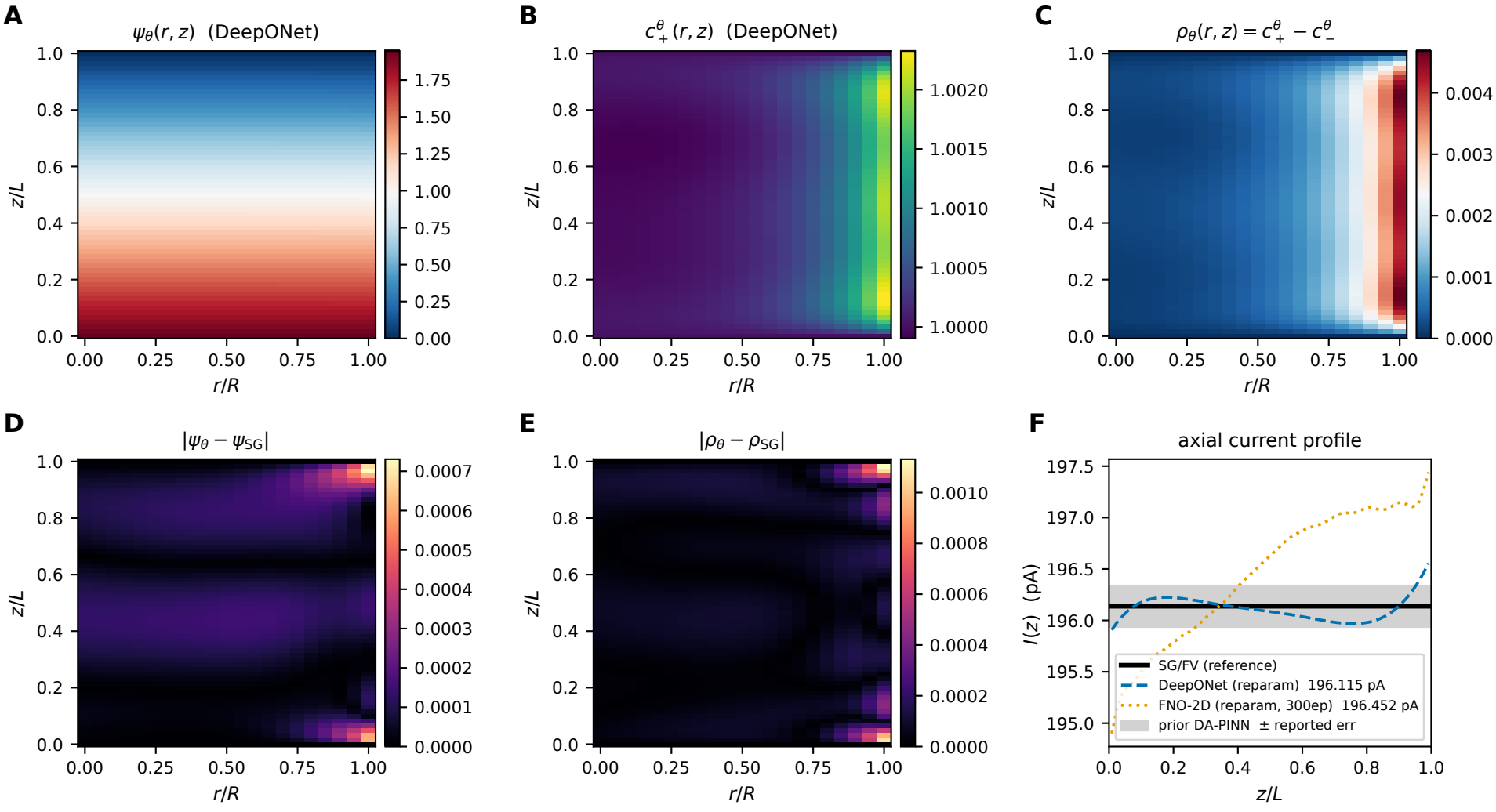

Benchmark state #0: $V_{app} = 50$ mV, $C_0 = 100$ mol m^{-3}, $q_w = -0.01$, $R = 5$ nm (out-of-sample)
A
$\psi_\theta(r,z)$ (DeepONet)
B
$c_+^\theta(r,z)$ (DeepONet)
C
$\rho_\theta(r,z) = c_+^\theta - c_-^\theta$
D
$|\psi_\theta - \psi_{SG}|$
E
$|\rho_\theta - \rho_{SG}|$
F
axial current profile
z/L
r/R
I(z) (pA)
SG/FV (reference)
DeepONet (reparam) 196.115 pA
FNO-2D (reparam, 300ep) 196.452 pA
prior DA-PINN ± reported err